Whole-Tumor Segmentation in Multimodal Brain MRI with a Patch-Based 3D U-Net

ENG 2440 Medical Imaging & AI in Healthcare — Assignment 2 · Brandon-Jason Adebayo (25011100)
Code, executed notebook and environment: https://github.com/B4ryterx/ENG2440-Assignment2

1. Task definition (A1)

The goal is to segment the **whole tumor** (edema + non-enhancing + enhancing tumor, i.e. every non-zero reference label) in the MSD Task01_BrainTumour data [1,2]. The **input** is one co-registered 4-channel 3D MRI volume per patient (FLAIR, T1, T1ce, T2; tensor [C=4, D, H, W]). The **output** is a voxel-wise tumor probability, thresholded at 0.5 into a binary mask on the native grid. The network decides per **voxel** (the smallest 1×1×1 mm element), but the **unit of analysis** for splitting and reporting is the **patient case** (four MRI volumes plus one label volume), because voxels from the same patient are not independent. A **slice** is a 2D cross-section (axial, coronal or sagittal); a **volume** is the full 240×240×155 scan. The intended use is technical and educational: a research segmentation, not a clinical tool, and the target is a human reference annotation, not pathology-confirmed tumor extent.

2. Volumes, geometry and case-level split (A2–A4, B)

Loading BRATS_001/003/004 gave image arrays of 240×240×155×4 (X, Y, Z, channel) and labels of 240×240×155. The channel order from *dataset.json* is 0 FLAIR, 1 T1w, 2 T1gd (T1ce), 3 T2w. The affine is the identity, the orientation codes are (R, A, S), the spacing is 1.0×1.0×1.0 mm (the 4th header zoom is the channel axis), and the label values are {0, 1, 2, 3} (background, edema, non-enhancing, enhancing). A header check of all 484 cases found a single grid: every image and label shares the same shape, affine, 1 mm spacing and RAS orientation. Fig. 1 shows the four channels of BRATS_001 at one physical location, the tumor centroid, voxel (157, 116, 67) = world (157, 116, 67) mm. All channels sit in one 4D NIfTI, so they share one affine by construction.

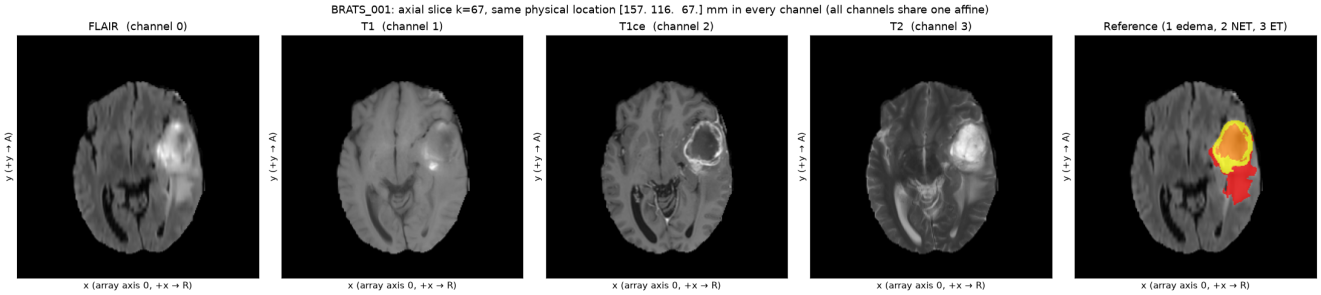


Fig. 1. BRATS_001, axial slice k = 67 through the tumor centroid: FLAIR, T1, T1ce, T2 and the multi-class reference overlaid on FLAIR (displayed with origin='lower' so +x→R, +y→A).

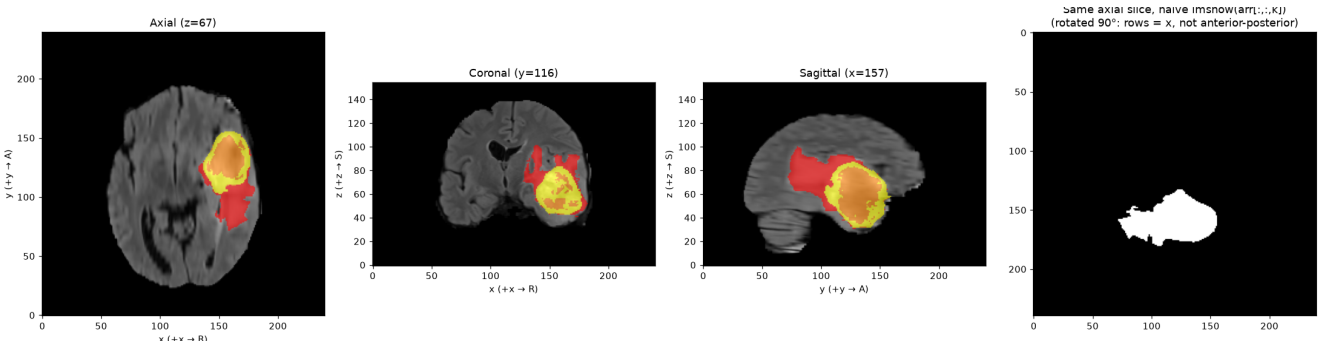


Fig. 2. Reference in axial, coronal and sagittal planes through the centroid, with anatomical axis directions taken from the affine. Right: the same axial slice drawn naively in raw index order, which appears rotated by 90°.

Why index order is insufficient. An array index (i, j, k) carries no anatomy; the affine defines which direction each axis points (RAS here, LPS or LAS elsewhere), the spacing in mm and the origin. Without it, left and right can silently swap, distances and volumes are wrong on anisotropic grids, and naive plotting rotates the anatomy (Fig. 2, right). Equal array shapes also do not prove that modalities and labels are aligned. Here index equals world mm, but that was *verified*, not assumed.

Split (B). The supplied *assignment2_split.csv* was used unchanged. Every case_id occurs once and all 968 image and label files exist. The train, validation and test sets are disjoint (all pairwise intersections are 0), so no patient's slices or patches can cross splits. Table 1 gives counts and reference tumor volumes (voxel count × voxel volume). The test volumes were computed only after the model, threshold and postprocessing were frozen (Section 6). The distributions are similar, with about a 30-fold range in size; the largest test tumor (318 mL) exceeds every training case.

Table 1. Whole-tumor reference volume per split.

Split	Cases	Mean ± SD (mL)	Median	Min – Max
Train	338	103.0 ± 57.0	95.5	8.5 – 265.3
Validation	73	100.8 ± 63.0	91.6	7.3 – 266.9
Test (read only in §6)	73	107.4 ± 66.3	89.1	10.1 – 318.3

3. Safe preprocessing (C)

Geometry and interpolation. Because all 484 cases already share one 1 mm RAS grid, **no resampling was applied**; the target grid is the native 240×240×155 grid, and all metrics are computed on it. To show correct practice, one case was resampled to 1.5 mm: linear interpolation for MRI, nearest neighbour for the label. Nearest neighbour kept exactly {0, 1, 2, 3} and preserved tumor volume (111.7 vs 110.6 mL). Linear interpolation of the label produced 301 distinct values, including fractions and fake class-2 voxels on edema/enhancing boundaries.

Normalisation. Each channel of each case was z-scored with the mean and SD of **brain voxels only** (non-zero in any channel), keeping the background at exactly 0. Including the zero background (83% of the array) drags the FLAIR mean from 516 to 86 and inflates the SD, which scales intensities by head size rather than tissue contrast. After normalisation, across all 338 training cases the per-case brain mean was 0.000 ± 0.002 and the SD 1.000 ± 0.002 for FLAIR and T2, and 0.999 ± 0.004 for T1 and T1ce (small deviations come from float16 storage).

Crop/pad and augmentation. Every volume is cropped to its brain bounding box plus 4 voxels. The box is computed from the image only, so it applies identically to test data, to all channels and to the label. It is then zero-padded to at least the 64^3 patch size (no case needed padding). This removes 59% of voxels, all of them background (median crop 146×177×143), and the box is stored so predictions are pasted back into the native grid. Training-only augmentation (Fig. 3): (i) left–right flip, p = 0.5, because the brain is roughly symmetric and tumors occur in both hemispheres; a flip is an exact voxel permutation, so the label stays binary. (ii) Per-channel intensity scale ×U(0.9, 1.1) and shift ±0.1 SD inside the brain, to mimic scanner variation without changing anatomy. Rotations and elastic warps (which need interpolation) and anterior-posterior or superior-inferior flips (anatomically impossible) were deliberately excluded. Validation and test data are never augmented.

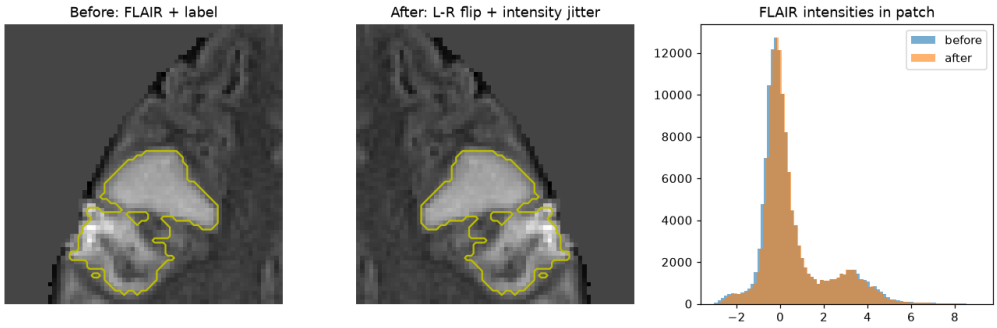


Fig. 3. One training patch before and after augmentation (FLAIR with label contour) and the intensity shift. The label after augmentation equals the flipped original exactly.

4. Patch-based volumetric training (D)

A PyTorch *Dataset* draws one random 64^3 patch per item from memory-mapped cached volumes. One printed batch had images [4, 4, 64, 64, 64] (float32) and masks [4, 1, 64, 64, 64] with values {0, 1}, and both shapes were asserted. Each item index has its own RNG seeded by (42, index), so both training runs visit the **same case sequence and augmentation draws**; only the patch-location rule differs. **Uniform** places the patch origin uniformly over the brain box. **Tumor-aware** centres the patch on a random tumor voxel (±16-voxel jitter) with p = 0.5, and otherwise samples uniformly. The model is a 4-level 3D U-Net [3] (widths 16–32–64–128, InstanceNorm + LeakyReLU, strided-convolution down-sampling, transposed-convolution up-sampling, concatenated skip connections) with **4 input channels, 1 output channel (logit) and 1,445,681 parameters**. Both runs start from the same saved initial weights (Table 2).

Table 2. Controlled training setup (identical for both samplers except the last two rows).

Item	Setting
Loss	BCE-with-logits + soft Dice (batch-level), equal weights
Optimiser / LR	Adam, lr 1e-3, poly decay (1 – t/T)^0.9, gradient-norm clip 5
Budget / seed	2,000 steps × batch 4 = 8,000 patches of 64^3; seed 42; Apple M4 Max (MPS)
Monitoring	every 200 steps: 64 fixed uniform validation patches (loss) + 8 full validation volumes (Dice); best-Dice checkpoint kept
Patches containing tumor	uniform 74.5% · tumor-aware 87.5%
Mean tumor-voxel fraction	uniform 6.8% · tumor-aware 14.1% (2.1×)

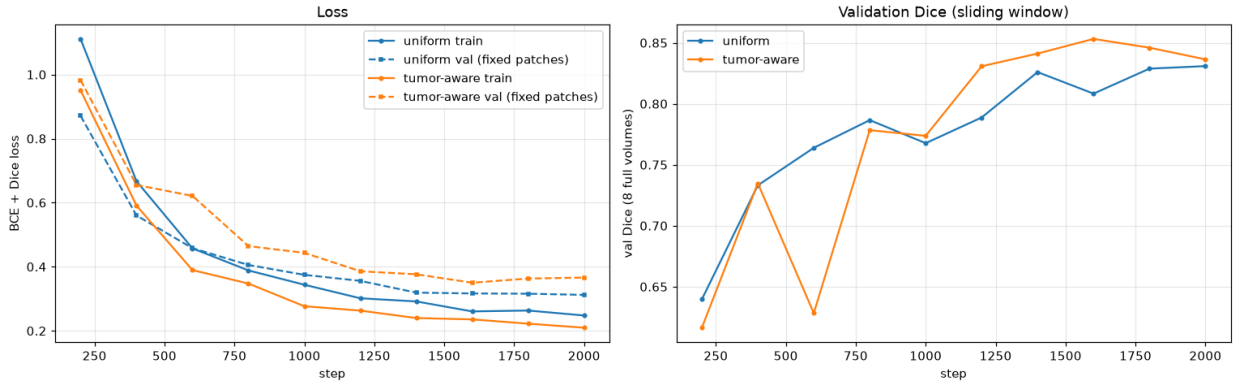


Fig. 4. Training and validation loss (left) and validation Dice on 8 full volumes (right) for both samplers. Training took 37.4 min (uniform) and 33.3 min (tumor-aware).

Curve diagnosis (validation only). Both training losses fall almost monotonically (1.11→0.25 and 0.95→0.21) and validation loss follows, with no widening gap, so there is **no clear overfitting**. Uniform validation Dice is still rising at step 2000 (0.831), so the budget leaves the model **mildly underfitted**. The tumor-aware run shows **instability** at step 600: Dice drops from 0.735 to 0.628 while training loss keeps falling. This happens early, at high learning rate, with tumor-dominated batches that briefly make the model over-call tumor; it recovers by step 800. Tumor-aware Dice peaks at step 1600 (0.853), then eases to 0.836 as its validation loss rises (0.350→0.366), the first sign of over-fitting to tumor-rich patches; the best-validation checkpoint avoids it.

5. Controlled sampling comparison (E)

Both conditions share the architecture, initial weights, split, loss, optimiser, budget, case sequence and checkpoint rule. Table 3 evaluates each best checkpoint on **all 73 validation volumes** with full sliding-window reconstruction.

Table 3. Sampling comparison on validation (no postprocessing).

Validation (n = 73)	Dice mean ± SD (median)	Sensitivity	FP vol. mean / median (mL)	FN vol. (mL)	HD95 median (mm)
Uniform	0.853 ± 0.087 (0.879)	0.879	14.6 / 10.1	11.7	6.3
Tumor-aware	0.856 ± 0.090 (0.884)	0.890	14.6 / 10.6	10.6	5.4

Mechanism. Tumor-aware sampling doubles the tumor and tumor-boundary voxels per step, giving the positive class and the edema/background border more gradient signal. The result is higher sensitivity, about 1 mL less missed tumor and a tighter boundary (median HD95 6.3→5.4 mm). The cost is fewer pure-background examples, so median false-positive volume rises slightly (10.1→10.6 mL). The gap is small because with 64 mm patches and ~100 mL tumors, 75% of uniform patches already contain tumor, and the Dice term already offsets class imbalance. The paired difference is +0.003 Dice (better in 44/73 cases, sign test $p \approx 0.10$): consistent but not conclusive for a single seed.

Selection. On validation evidence alone, **tumor-aware sampling (p = 0.5, best checkpoint)** was selected: highest mean Dice, higher sensitivity, lower missed volume and HD95, equal mean false-positive volume. **Disadvantage of excessive oversampling:** the network learns the tumor prior of the sampler, not of the brain. At inference, where >90% of windows are background, it over-calls FLAIR-bright non-tumor tissue, inflates volumes and is poorly calibrated, and background hard negatives are under-sampled. A hint of this is already visible at p = 0.5: a higher validation loss on uniformly sampled patches (0.366 vs 0.312).

6. Full-volume reconstruction and test evaluation (F)

Sliding window. Each cropped volume is covered by 64^3 windows with **50% overlap (stride 32)**, and the last window is aligned to the volume end (e.g. 80 windows for a 150×184×150 crop). Each window's sigmoid probabilities are multiplied by a Gaussian importance map ($\sigma = 8$ voxels, centre weight 1). The weighted probabilities are summed and divided by the summed weights; only then is the map thresholded at 0.5 and pasted into the native grid. HD95 is the 95th percentile of the pooled symmetric surface distances, computed with the header spacing in mm; volume = voxel count × spacing product / 1000 mL.

Postprocessing (decided before testing). Remove 26-connected components smaller than 0.5 mL. *Rationale:* whole tumor is a large region, while isolated specks are mostly false positives (vessels, FLAIR artefacts, window-border noise), and a size threshold, unlike keep-largest, preserves multifocal lesions. *Decision on validation:* Dice 0.856→0.859, mean HD95 19.4→17.7 mm, false-positive volume 14.6→14.2 mL, sensitivity unchanged, so the rule was **kept**. The test results below are shown before and after for transparency only; they played no part in the decision.

Table 4. Untouched test split, case-wise metrics as mean ± SD (median). Per-case values are in Appendix A.

Test (n = 73)	Dice	IoU	Sensitivity	HD95 (mm)	Empty
Before postproc.	0.879 ± 0.077 (0.896)	0.792 ± 0.110 (0.811)	0.891 ± 0.094 (0.916)	8.59 ± 11.19 (4.47)	0
After (final)	0.881 ± 0.077 (0.896)	0.794 ± 0.110 (0.812)	0.890 ± 0.095 (0.916)	8.21 ± 10.80 (4.24)	0

Postprocessing barely changes overlap but improves HD95 (median 4.47→4.24 mm): a few distant false-positive voxels hardly affect Dice but dominate the 95th-percentile surface distance. No prediction was empty. **Volume:** the final absolute

volume error is 12.7 ± 13.2 mL (median 7.9 mL), and the signed percentage error is +2.6 ± 19.0% (median +1.1%), so there is little systematic bias but the relative error is largest for small tumors (Fig. 5).

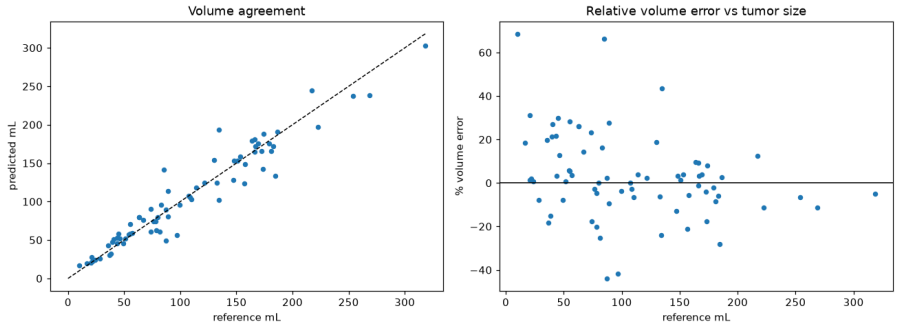


Fig. 5. Predicted vs reference tumor volume (left) and percentage volume error vs tumor size (right), test set.

7. Three-plane error analysis (G)

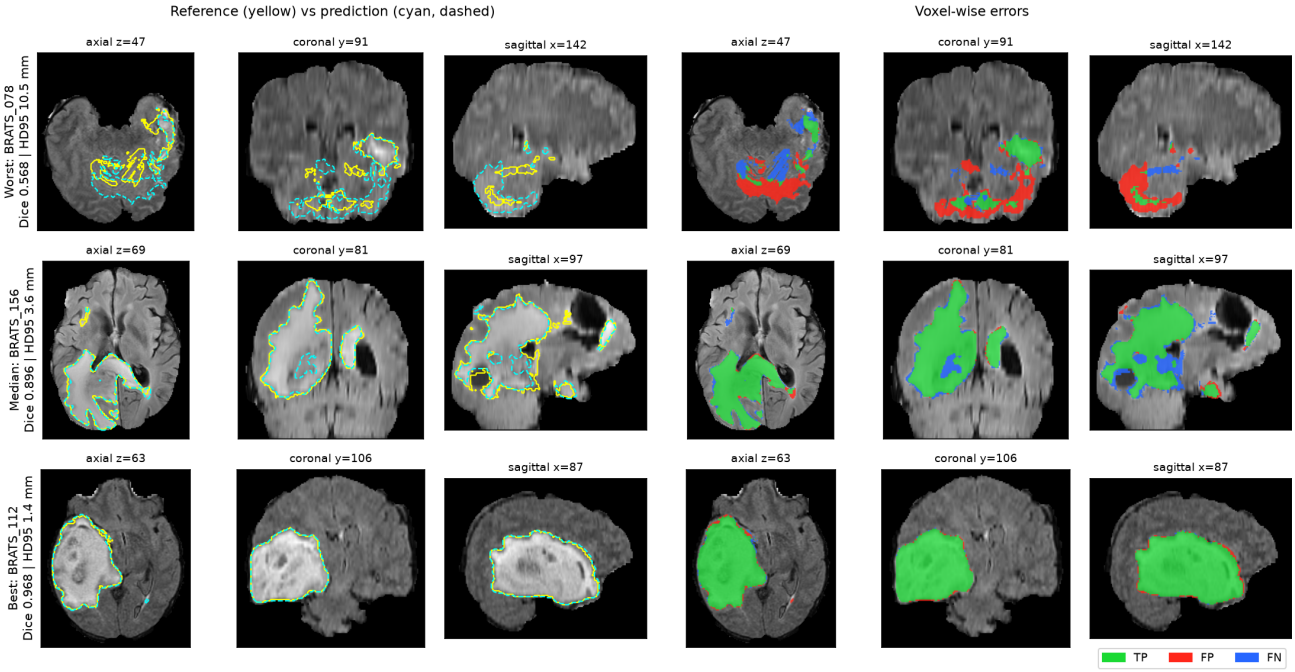


Fig. 6. Worst, median and best test cases by Dice: axial, coronal and sagittal planes through the lesion, cropped to the brain. **Left three columns:** reference (yellow) and prediction (cyan, dashed) contours on FLAIR. **Right three columns:** true positives green, false positives red, false negatives blue. The full-size versions are in the notebook (G1, G2).

Best case (Dice 0.968): a large, compact lesion. 93% of false-positive and 98% of false-negative voxels lie within 2 mm of the reference boundary, so the only error is a one-voxel rim of **boundary disagreement**, comparable to inter-rater variation. **Median case** (Dice 0.896): the errors are again near the boundary (94% of false positives within 2 mm), but the case is under-segmented, missing 34.7 mL versus 8.9 mL of false positives. The missed regions surround dark cavity and necrotic tissue next to the ventricle, which locally does not look like bright edema. **Worst case** (Dice 0.568) fails differently: only 28% of its 76.6 mL of false positives touch the reference boundary, spread over 107 components in the posterior fossa and cerebellum. A 64 mm window there contains no tumor and no contralateral tissue to compare with, so FLAIR brightness typical of the skull base is mistaken for edema. This is a **patch-context** failure, made worse because infratentorial tumors are rare in training. **Small structures** are also harder: Dice correlates with tumor size (Spearman $\rho = 0.33$, $p = 0.004$), and the smallest test tumor (10.1 mL) scores Dice 0.60 with HD95 33.6 mm.

Table 5. Overlap, boundary and volume metrics for the three cases in Fig. 6.

Case	Dice	HD95 (mm)	Ref. / pred. (mL)	Volume error
Worst BRATS_078	0.568	10.5	85.1 / 140.7	+55.6 mL (+65%)
Median BRATS_156	0.896	3.6	222.6 / 196.8	–25.8 mL (–12%)
Best BRATS_112	0.968	1.4	169.2 / 175.7	+6.5 mL (+4%)

One metric is not enough. The metrics agree on these three cases, but over all test cases they are only partly related (Spearman: Dice–HD95 −0.74, Dice–|volume error| −0.52), and each misses failures the others catch. Dice ignores a tiny distant false-positive island (BRATS_467: Dice 0.93 but HD95 59 mm) and penalises small tumors. HD95 can explode from one speck while saying nothing about how much tissue is wrong. Volume error can be small while false positives and false negatives cancel: the median case's net −25.8 mL hides 43.6 mL of misclassified tissue. Overlap, boundary and volume metrics must therefore be reported together [5].

8. Limitations

(1) **Small training budget:** 2,000 steps (8,000 patches, about 35 min per run), one seed and no repeated runs. Validation Dice was still improving, and the +0.003 sampler difference is within run-to-run noise. (2) **No external-site validation:** all splits come from the same skull-stripped, co-registered, 1 mm-resampled BraTS 2016–17 collection, so performance on raw clinical scans from other scanners, field strengths, protocols or pathologies is unknown and probably lower. (3) **Reference standard:** labels are human annotations with known inter-rater variability at the edema boundary, not pathology, so a Dice around 0.9 is near the agreement ceiling. (4) **Context and target:** 64 mm patches limit global context (BRATS_078), and only binary whole tumor was modelled, not the clinically important tumor core or enhancing tumor. (5) **Minor protocol bias:** the 8 monitoring cases used for checkpoint selection are also part of the 73-case validation comparison, and Apple GPU (MPS) kernels are not bit-wise deterministic.

References

[1] M. Antonelli et al., "The Medical Segmentation Decathlon," *Nature Communications*, 13:4128, 2022. [2] B. H. Menze et al., "The multimodal brain tumor image segmentation benchmark (BRATS)," *IEEE TMI*, 34(10):1993–2024, 2015; MSD Task01_BrainTumour, medicaldecathlon.com, accessed 2026. [3] Ö. Çiçek et al., "3D U-Net: learning dense volumetric segmentation from sparse annotation," MICCAI 2016. [4] F. Isensee et al., "nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods*, 18:203–211, 2021 (design ideas: foreground oversampling, Gaussian sliding-window blending, poly learning rate; re-implemented, no code copied). [5] L. Maier-Hein et al., "Metrics reloaded: recommendations for image analysis validation," *Nature Methods*, 21:195–212, 2024. [6] M. Havaei et al., "HeMIS: Hetero-modal image segmentation," MICCAI 2016. [7] A. Paszke et al., "PyTorch: an imperative style, high-performance deep learning library," NeurIPS 2019.

Appendix A — Per-case test results (final configuration, after postprocessing)

Case	Dice	IoU	Sens.	HD95	Ref mL	Pred mL	|Err| mL	Err %	Case	Dice	IoU	Sens.	HD95	Ref mL	Pred mL	|Err| mL	Err %
B002	0.876	0.779	0.989	3.6	63.1	79.4	16.3	+25.8	B252	0.922	0.856	0.923	4.6	79.9	80.0	0.1	+0.2
B012	0.918	0.848	0.923	3.3	21.6	21.8	0.2	+1.1	B253	0.878	0.783	0.902	12.6	54.4	57.3	2.9	+5.3
B016	0.895	0.810	0.967	3.7	82.7	96.1	13.4	+16.1	B260	0.955	0.914	0.942	2.0	108.4	105.5	2.9	-2.7
B033	0.947	0.900	0.966	1.7	114.0	118.4	4.4	+3.9	B262	0.913	0.840	0.916	2.0	23.9	24.1	0.1	+0.6
B039	0.867	0.765	0.787	3.6	37.0	30.2	6.8	-18.4	B281	0.898	0.815	0.932	7.0	173.9	187.3	13.3	+7.7
B050	0.941	0.888	0.928	2.2	76.2	74.2	2.0	-2.6	B283	0.914	0.842	0.970	4.2	217.4	244.0	26.7	+12.3
B059	0.943	0.892	0.925	2.2	99.6	95.9	3.7	-3.7	B293	0.876	0.779	0.989	3.6	63.1	79.4	16.3	+25.8
B065	0.873	0.775	0.912	5.0	166.1	180.7	14.6	+8.8	B303	0.892	0.805	0.865	4.9	183.1	172.1	10.9	-6.0
B067	0.924	0.858	0.948	12.3	55.3	58.2	3.0	+5.4	B305	0.861	0.756	0.973	10.8	55.1	69.4	14.3	+25.9
B078	0.568	0.397	0.753	10.5	85.1	140.7	55.6	+65.3	B307	0.877	0.781	0.924	3.2	46.3	51.3	5.0	+10.8
B079	0.906	0.828	0.863	2.2	89.1	80.6	8.5	-9.5	B322	0.822	0.698	0.941	49.2	20.9	27.0	6.1	+29.0
B088	0.917	0.846	0.887	3.3	253.7	237.2	16.5	-6.5	B341	0.874	0.776	0.844	17.5	132.7	123.6	9.1	-6.8
B097	0.958	0.919	0.976	1.4	153.2	158.8	5.7	+3.7	B354	0.920	0.851	0.930	4.0	87.3	89.2	1.9	+2.2
B108	0.862	0.758	0.871	3.2	121.8	124.3	2.5	+2.0	B366	0.770	0.626	0.877	9.2	89.2	113.9	24.7	+27.7
B112	0.968	0.939	0.987	1.4	169.2	175.7	6.5	+3.8	B372	0.791	0.654	0.871	5.0	39.4	47.4	7.9	+20.1
B131	0.820	0.695	0.747	9.4	173.3	142.7	30.7	-17.7	B374	0.863	0.759	0.952	5.4	43.5	52.4	8.9	+20.5
B137	0.909	0.833	0.850	4.9	147.0	128.0	19.0	-12.9	B376	0.603	0.432	0.770	33.6	10.1	15.7	5.6	+55.5
B141	0.888	0.799	0.836	5.4	268.7	237.1	31.7	-11.8	B377	0.887	0.797	0.987	2.8	73.7	90.3	16.6	+22.5
B142	0.960	0.923	0.954	2.0	166.2	164.3	1.9	-1.1	B387	0.937	0.882	0.943	2.4	150.8	152.5	1.7	+1.1
B144	0.922	0.856	0.926	2.4	51.3	51.7	0.4	+0.8	B396	0.946	0.898	0.915	1.4	110.2	102.9	7.2	-6.6
B156	0.896	0.812	0.844	3.6	222.6	196.8	25.8	-11.6	B399	0.949	0.902	0.994	2.0	163.6	179.3	15.7	+9.6
B158	0.937	0.882	0.896	2.2	49.2	44.8	4.4	-8.9	B401	0.869	0.768	0.999	30.7	44.8	58.2	13.4	+30.0
B163	0.871	0.771	0.782	5.1	78.4	62.4	15.9	-20.3	B412	0.924	0.858	0.904	5.0	172.7	165.4	7.3	-4.2
B168	0.818	0.692	0.920	6.0	40.5	50.5	10.1	+24.9	B414	0.861	0.756	0.770	8.4	156.8	123.8	33.0	-21.1
B169	0.692	0.529	0.539	10.8	87.4	48.7	38.7	-44.2	B436	0.947	0.900	0.958	1.4	43.9	44.8	1.0	+2.2
B171	0.929	0.868	0.936	1.4	20.6	20.9	0.3	+1.3	B437	0.904	0.825	0.988	3.3	129.8	153.9	24.0	+18.5
B173	0.958	0.920	0.947	1.4	179.3	175.1	4.2	-2.3	B440	0.896	0.812	0.959	2.0	66.8	76.2	9.4	+14.0
B188	0.763	0.617	0.828	19.2	35.8	41.9	6.0	+16.9	B447	0.956	0.915	0.956	1.7	107.2	107.3	0.1	+0.1
B192	0.699	0.538	0.552	10.3	96.8	56.1	40.7	-42.1	B451	0.964	0.930	0.976	1.4	186.3	191.2	4.9	+2.6
B198	0.948	0.902	0.908	2.4	181.0	165.5	15.5	-8.6	B453	0.841	0.725	0.740	7.3	134.3	102.0	32.3	-24.0
B222	0.919	0.850	0.889	5.4	78.3	73.2	5.1	-6.5	B461	0.950	0.905	0.964	2.0	148.0	152.4	4.3	+2.9
B230	0.863	0.759	0.829	3.0	28.3	26.0	2.2	-7.9	B462	0.818	0.692	0.756	6.6	38.3	32.5	5.8	-15.2
B241	0.807	0.676	0.980	12.8	134.6	192.3	57.8	+42.9	B466	0.901	0.820	0.979	2.0	16.7	19.6	2.9	+17.3
B246	0.842	0.728	0.736	32.0	81.4	60.8	20.6	-25.3	B467	0.935	0.878	0.906	59.4	157.7	148.0	9.8	-6.2
B247	0.885	0.794	0.806	23.2	74.0	60.7	13.3	-17.9	B481	0.872	0.774	0.888	31.1	57.2	59.2	2.1	+3.6
B249	0.932	0.873	0.910	7.1	318.3	302.7	15.6	-4.9	B482	0.788	0.650	0.676	10.3	184.7	132.5	52.2	-28.3
B250	0.892	0.805	0.903	9.0	166.6	170.9	4.3	+2.6									

HD95 in mm (pooled symmetric surface distances, 1 mm spacing). Volume error = predicted − reference. Summary: Dice 0.881 ± 0.077 (0.896), IoU 0.794 ± 0.110 (0.812), sensitivity 0.890 ± 0.095 (0.916), HD95 8.21 ± 10.80 (4.24) mm, |volume error| 12.7 ± 13.2 (7.9) mL (mean ± SD (median)); empty predictions: 0.

Appendix B — Bonus questions

B1. Missing-modality stress test (same final model, no retraining, postprocessing as in the final configuration).

Test input	Dice mean	HD95 median (mm)	|Vol. err.| mean (mL)	Vol. err. median	Empty
All 4 channels	0.881	4.2	12.7	+1.1%	0
FLAIR = 0	0.366	25.4	72.1	–71.0%	6
T1 = 0	0.801	25.2	32.0	+30.7%	0
T1ce = 0	0.849	5.9	18.7	–1.4%	0
T2 = 0	0.847	5.4	19.4	–3.8%	0

FLAIR is by far the most influential channel. Zeroing it collapses Dice from 0.881 to 0.366, produces 6 empty predictions, raises median HD95 from 4.2 to 25.4 mm and shrinks volumes (median −71%). Whole tumor is defined mostly by peritumoral edema, which FLAIR shows best. Zeroing T1 is second (Dice 0.801) but fails the other way: volumes are over-estimated (+31%) and median HD95 reaches 25 mm as distant tissue is labelled tumor. Removing T1ce or T2 costs only about 0.03 Dice, because T2 largely duplicates FLAIR's edema contrast and T1ce mainly informs the enhancing core. Zeroing is an imperfect simulation of a missing scan: after z-scoring, 0 means *average brain intensity*, so the network receives a flat, average-tissue image it never saw in training. That is an out-of-distribution input rather than an explicit "unknown", and the model cannot tell the channel is absent. Real missing acquisitions may instead be substituted by other protocols, degraded scans or synthetic images. Training could be redesigned with **modality dropout** (randomly blanking channels plus binary availability-indicator channels), per-modality encoders fused by order-independent statistics (HeMIS [6]), or distillation from a full-modality teacher, and then validated on genuinely incomplete cases.

B2. Sliding-window seam test (same final model; runtime includes metric computation).

Overlap	Stride	Windows (median case)	Runtime, 73 cases (min)	Dice mean	HD95 median (mm)	|Vol. err.| (mL)
0%	64	27	1.6	0.859	5.0	14.8
25%	48	36	2.0	0.873	5.1	13.3
50%	32	80	3.8	0.881	4.2	12.7

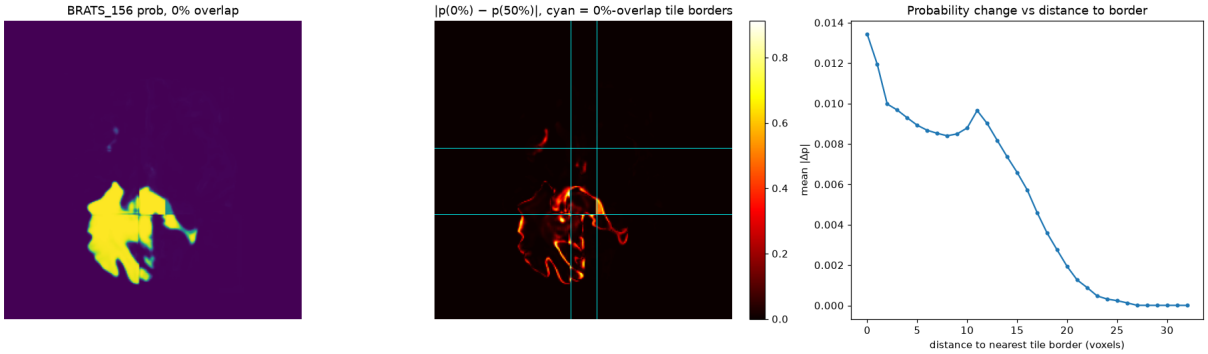


Fig. B1. BRATS_156: probability map at 0% overlap with visible tile seams (left); |p(0%) – p(50%)| with 0%-overlap tile borders (middle); mean |Δp| vs distance to the nearest tile border (right).

More overlap steadily improves every metric (Dice 0.859→0.873→0.881; |volume error| 14.8→13.3→12.7 mL) at about 2.3× the runtime of 0% (27→80 windows). At 0% overlap the probability map shows straight discontinuities along tile borders, and the change between 0% and 50% overlap is largest exactly at the borders (mean |Δp| 0.013) and decays into the tile interior. Voxels near a window edge are predicted with truncated context and the network's zero padding; with overlap, every voxel is also seen from a window where it is central, and Gaussian blending down-weights the unreliable edges. Overlap therefore helps most when structures cross tile borders and are large relative to the patch, which is almost always true here (64 mm windows, ~100 mL tumors), and it matters less when objects sit well inside a single tile. Choosing the overlap after repeatedly inspecting test labels would turn the test set into a tuning set: the chosen setting would partly fit test-specific noise and the reported score would be optimistically biased, no longer an independent estimate of generalisation. Here, 50% overlap was fixed before testing and this comparison is reported post hoc; any such choice should be made on validation data.